

Domain Microcosm Blueprint — HealthTech

DSBA 6171: Data Engineering for AI | Medical Claims & Payer Auditing

1. Part 3: Unstructured Knowledge Corpus Design

The HealthTech knowledge corpus contains 7 distinct documents bridging medical coverage policies, procedure allowances, statutory requirements, and claims auditing procedures:

- **HT-POL-001_v2026.pdf (PDF):** 2026 Inpatient & Outpatient Coverage Policy (Hierarchical structure, *CURRENT*).
- **HT-POL-001_v2024.pdf (PDF):** 2024 Legacy Policy (*SUPERSEDED* state for version diversity).
- **HT-GUIDE-002.pdf (PDF):** Procedure Allowance & Fee Schedule Matrix (Contains substantial data table).
- **HT-PROC-001.md (Markdown):** High-Dollar Claim Audit & Escalation Workflow (Standard operating procedure).
- **HT-REG-001.txt (Text):** Federal Payer Compliance & Prompt Pay Standards (Public regulatory guidance).
- **HT-MEMO-001.md (Markdown):** Temporary Emergency Room Copay Waiver Directive (Labeled *SYNTHETIC INTERNAL*).
- **HT-FAQ-001.txt (Text):** Examiner Audit & Coding Desk FAQ (Operational query guidance).

2. Part 4: Document Metadata Inventory Table

Below is the complete document metadata control layer saved in `/metadata/document_metadata.csv`:

doc_id	title	doc_type	owner	eff_date	ver	status	access_group
HT-POL-001	Inpatient & Outpatient Policy	Policy	Medical Policy	2026-01-01	v2.0	current	Claims_Auditors
HT-POL-001-OLD	Inpatient & Outpatient Policy (Legacy)	Policy	Medical Policy	2024-01-01	v1.0	superseded	Claims_Auditors
HT-GUIDE-002	Procedure Allowance Matrix	Guideline	Provider Relations	2025-07-01	v1.3	current	Billing_Ops
HT-PROC-001	Claim Audit & Escalation Workflow	Procedure	Claims Ops	2025-10-01	v1.0	approved	Senior_Auditors
HT-REG-001	Payer Compliance Standards	Regulation	Legal Compliance	2023-05-15	v1.0	current	All_Employees
HT-MEMO-001	Temporary ER Copay Waiver	Memo	Executive Ops	2026-02-01	v1.0-DRAFT	draft	Executive_Auditors
HT-FAQ-001	Examiner Audit & Desk FAQ	FAQ	Claims Ops	2025-11-15	v1.1	current	Billing_Ops

3. Linkage to Structured Data & Microcosm Structure

Key Structured-to-Unstructured Linkages:

- `claims_ledger.csv (procedure_code) + patient_accounts.csv (plan_code)` → Maps to HT-GUIDE-002.pdf to verify allowed amounts and pre-authorization necessity.
- `claims_ledger.csv (billed_amount > $25,000)` → Triggers audit escalation workflow in HT-PROC-001.md.